

CSA0503-DATABASE MANAGEMENT SYSTEMS

CO3 AT3

Case Study

1. Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.

SELECT

O.OrderID,
C.CustID,
C.CustName,
P.ProdID,
P.ProdName,
I.Qty,
P.Price

FROM 'ORDER' O

JOIN CUSTOMER C ON O.CustID = C.CustID

JOIN ORDER_ITEM I ON O.OrderID = I.OrderID

JOIN PRODUCT P ON I.ProdID = P.ProdID;

```
mysql> SELECT
-> O.OrderID,
-> C.CustID,
-> C.CustName,
-> P.ProdID,
-> P.ProdName,
-> I.Qty,
-> P.Price
-> FROM 'ORDER' O
-> JOIN CUSTOMER C ON O.CustID = C.CustID
-> JOIN ORDER_ITEM I ON O.OrderID = I.OrderID
-> JOIN PRODUCT P ON I.ProdID = P.ProdID;
```

OrderID	CustID	CustName	ProdID	ProdName	Qty	Price
1001	101	Arun	201	Laptop	1	55000.00
1001	101	Arun	202	Mouse	2	800.00
1003	101	Arun	202	Mouse	1	800.00
1002	102	Divya	203	Keyboard	1	1500.00

4 rows in set (0.01 sec)

2. A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.

SELECT

P.PID,

P.PName,

D.DocID,

D.DocName,

D.Specialization,

W.WardNo,

W.WardName

FROM PATIENT P

JOIN DOCTOR D ON P.DocID = D.DocID

JOIN WARD W ON P.WardNo = W.WardNo; (102, 'Divya', 'C003', 'CN', 'A');

```
-> P.PID,
-> P.PName,
-> D.DocID,
-> D.DocName,
-> D.Specialization,
-> W.WardNo,
-> W.WardName
-> FROM PATIENT P
-> JOIN DOCTOR D ON P.DocID = D.DocID
-> JOIN WARD W ON P.WardNo = W.WardNo;
```

PID	PName	DocID	DocName	Specialization	WardNo	WardName
1001	Arun	101	Dr. Kumar	Cardiology	1	General Ward
1003	Karthik	101	Dr. Kumar	Cardiology	1	General Ward
1002	Divya	102	Dr. Priya	Neurology	2	ICU
1004	Meena	103	Dr. Ravi	Orthopedics	3	Emergency Ward

4 rows in set (0.01 sec)

3. Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.

SELECT

S.SID,

S.SName,

C.CourseID,

C.CourseName,

C.Faculty,

E.Grade

FROM ENROLLMENT E

JOIN STUDENT S ON E.SID = S.SID

JOIN COURSE C ON E.CourseID = C.CourseID;

```
mysql> SELECT
-> S.SID,
-> S.SName,
-> C.CourseID,
-> C.CourseName,
-> C.Faculty,
-> E.Grade
-> FROM ENROLLMENT E
-> JOIN STUDENT S ON E.SID = S.SID
-> JOIN COURSE C ON E.CourseID = C.CourseID;
```

SID	SName	CourseID	CourseName	Faculty	Grade
101	Arun	C001	Database Systems	Dr. Kumar	A
102	Divya	C001	Database Systems	Dr. Kumar	A+
101	Arun	C002	Data Structures	Dr. Priya	B+
103	Karthik	C002	Data Structures	Dr. Priya	A
102	Divya	C003	Operating Systems	Dr. Ravi	B
104	Meena	C003	Operating Systems	Dr. Ravi	A+

6 rows in set (0.00 sec)

4. Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.

SELECT

B.FlightNo,

B.Date,

P.PassID,

P.PassName,

B.Seat,

F.DeptCity,

F.ArrCity

FROM BOOKING B

JOIN PASSENGER P

ON B.PassID = P.PassID

JOIN FLIGHT F

ON B.FlightNo = F.FlightNo;

```
mysql> SELECT
-> B.FlightNo,
-> B.Date,
-> P.PassID,
-> P.PassName,
-> B.Seat,
-> F.DeptCity,
-> F.ArrCity
-> FROM BOOKING B
-> JOIN PASSENGER P
-> ON B.PassID = P.PassID
-> JOIN FLIGHT F
-> ON B.FlightNo = F.FlightNo;
```

FlightNo	Date	PassID	PassName	Seat	DeptCity	ArrCity
F101	2026-08-20	101	Arun	12A	Chennai	Delhi
F101	2026-08-20	102	Divya	12B	Chennai	Delhi
F102	2026-08-21	103	Karthik	10A	Mumbai	Bangalore
F103	2026-08-22	104	Meena	15C	Chennai	Hyderabad

4 rows in set (0.00 sec)

5. Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.

```
SELECT
    E.EmpID,
    E.EmpName,
    D.DeptName,
    M.ManagerName,
    C.CountryName
FROM EMPLOYEE E
JOIN DEPARTMENT D
    ON E.DeptID = D.DeptID
JOIN MANAGER M
    ON E.ManagerID = M.ManagerID
JOIN COUNTRY C
    ON E.CountryID = C.CountryID;
```

```
mysql> SELECT
->     E.EmpID,
->     E.EmpName,
->     D.DeptName,
->     M.ManagerName,
->     C.CountryName
-> FROM EMPLOYEE E
-> JOIN DEPARTMENT D
->     ON E.DeptID = D.DeptID
-> JOIN MANAGER M
->     ON E.ManagerID = M.ManagerID
-> JOIN COUNTRY C
->     ON E.CountryID = C.CountryID;
```

EmpID	EmpName	DeptName	ManagerName	CountryName
1001	Arun	Human Resources	Raj Kumar	India
1002	Divya	Finance	Priya Sharma	India
1003	Karthik	Information Technology	John Smith	USA
1005	Rahul	Information Technology	John Smith	India
1004	Meena	Marketing	David Wilson	UK

```
5 rows in set (0.00 sec)
```

6. Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.

```
SELECT
    M.MemberID,
    M.MemberName,
    B.BookID,
    B.Title,
    B.Author,
    B.Genre,
    BR.BorrowDate
FROM MEMBER M
JOIN BORROWING BR
    ON M.MemberID = BR.MemberID
JOIN BOOK B
    ON BR.BookID = B.BookID;
```

```
mysql> SELECT
->     M.MemberID,
->     M.MemberName,
->     B.BookID,
->     B.Title,
->     B.Author,
->     B.Genre,
->     BR.BorrowDate
-> FROM MEMBER M
-> JOIN BORROWING BR
->     ON M.MemberID = BR.MemberID
-> JOIN BOOK B
->     ON BR.BookID = B.BookID;
```

MemberID	MemberName	BookID	Title	Author	Genre	BorrowDate
101	Arun	201	Database Systems	Ramakrishnan	Database	2026-08-01
101	Arun	202	Data Structures	Mark Allen	Computer Science	2026-08-03
102	Divya	201	Database Systems	Ramakrishnan	Database	2026-08-05
102	Divya	203	Artificial Intelligence	Stuart Russell	AI	2026-08-07
103	Karthik	204	Operating Systems	Galvin	Computer Science	2026-08-10

```
5 rows in set (0.00 sec)
```

7. Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.

```
SELECT CallID, COUNT(*)  
FROM TELECOM_BILL  
GROUP BY CallID  
HAVING COUNT(*) > 1;
```

```
SELECT PhoneNo,  
       COUNT(DISTINCT CustomerID) AS Customers  
FROM TELECOM_BILL  
GROUP BY PhoneNo;
```

```
mysql> SELECT CustomerID,  
->         COUNT(DISTINCT CustomerName) AS Names,  
->         COUNT(DISTINCT PhoneNo) AS PhoneNumbers,  
->         COUNT(DISTINCT PlanID) AS Plans  
-> FROM TELECOM_BILL  
-> GROUP BY CustomerID;
```

CustomerID	Names	PhoneNumbers	Plans
101	1	1	1
102	1	1	1
103	1	1	1
104	1	1	1

4 rows in set (0.00 sec)

```
mysql> SELECT PlanID,  
->         COUNT(DISTINCT PlanName) AS PlanNames  
-> FROM TELECOM_BILL  
-> GROUP BY PlanID;
```

PlanID	PlanNames
P01	1
P02	1
P03	1

3 rows in set (0.00 sec)

```
mysql> SELECT PhoneNo,  
->         COUNT(DISTINCT CustomerID) AS Customers  
-> FROM TELECOM_BILL  
-> GROUP BY PhoneNo;
```

PhoneNo	Customers
9876500001	1
9876500002	1
9876500003	1
9876500004	1

8. A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.

SELECT

P.PID,

P.PName,

P.SupplierID,

S.SupplierName,

P.Category,

P.Price,

P.Warehouse

FROM PRODUCT P

JOIN SUPPLIER S

ON P.SupplierID = S.SupplierID;

```
mysql> SELECT
-> P.PID,
-> P.PName,
-> P.SupplierID,
-> S.SupplierName,
-> P.Category,
-> P.Price,
-> P.Warehouse
-> FROM PRODUCT P
-> JOIN SUPPLIER S
-> ON P.SupplierID = S.SupplierID;
```

PID	PName	SupplierID	SupplierName	Category	Price	Warehouse
1001	Laptop	101	ABC Suppliers	Electronics	55000.00	Chennai Warehouse
1002	Mouse	101	ABC Suppliers	Accessories	800.00	Chennai Warehouse
1003	Keyboard	102	Global Traders	Accessories	1500.00	Bangalore Warehouse
1005	Printer	102	Global Traders	Electronics	15000.00	Bangalore Warehouse
1004	Monitor	103	Prime Distributors	Electronics	12000.00	Chennai Warehouse

5 rows in set (0.00 sec)

9. Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.

```
SELECT
    S.StudentID,
    S.StudentName,
    C.ClassName,
    T.TeacherName,
    SUB.SubjectName,
    R.Marks,
    R.Grade
FROM RESULT R
JOIN STUDENT S
    ON R.StudentID = S.StudentID
JOIN CLASS C
    ON S.ClassID = C.ClassID
JOIN TEACHER T
    ON C.TeacherID = T.TeacherID
JOIN SUBJECT SUB
    ON R.SubjectID = SUB.SubjectID;
```

```
mysql> SELECT
->     S.StudentID,
->     S.StudentName,
->     C.ClassName,
->     T.TeacherName,
->     SUB.SubjectName,
->     R.Marks,
->     R.Grade
-> FROM RESULT R
-> JOIN STUDENT S
->     ON R.StudentID = S.StudentID
-> JOIN CLASS C
->     ON S.ClassID = C.ClassID
-> JOIN TEACHER T
->     ON C.TeacherID = T.TeacherID
-> JOIN SUBJECT SUB
->     ON R.SubjectID = SUB.SubjectID;
```

StudentID	StudentName	ClassName	TeacherName	SubjectName	Marks	Grade
1001	Arun	10-A	Ravi Kumar	Mathematics	85	A
1001	Arun	10-A	Ravi Kumar	Computer Science	90	A+
1002	Divya	10-A	Ravi Kumar	Mathematics	78	B+
1002	Divya	10-A	Ravi Kumar	Physics	82	A
1003	Karthik	10-B	Priya Sharma	Computer Science	88	A
1004	Meena	10-B	Priya Sharma	English	75	B+
1005	Rahul	11-A	John Peter	Mathematics	92	A+

```
7 rows in set (0.00 sec)
```

10. Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.

```
SELECT
    UM.UserID,
    UM.MovieID,
    UP.PlatformID
FROM USER_MOVIE UM
JOIN USER_PLATFORM UP
    ON UM.UserID = UP.UserID
JOIN MOVIE_PLATFORM MP
    ON UM.MovieID = MP.MovieID
    AND UP.PlatformID = MP.PlatformID;
```

```
mysql> SELECT
->     UM.UserID,
->     UM.MovieID,
->     UP.PlatformID
-> FROM USER_MOVIE UM
-> JOIN USER_PLATFORM UP
->     ON UM.UserID = UP.UserID
-> JOIN MOVIE_PLATFORM MP
->     ON UM.MovieID = MP.MovieID
->     AND UP.PlatformID = MP.PlatformID;
+-----+-----+-----+
| UserID | MovieID | PlatformID |
+-----+-----+-----+
|      1 |      101 |         1001 |
|      1 |      102 |         1001 |
|      2 |      101 |         1001 |
|      2 |      102 |         1001 |
|      3 |      103 |         1002 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```